Attitudes towards information technology in Indian English and German novels since 2000

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Introduction

In 2020, the median age in India was 27, while in the same year the median age in Germany was 44.9. [1] It is reasonable to assume that this difference affects other aspects. Surveys on German attitudes towards information technology, for example, show a lot of skepticism and fears (Künsting 2018, Eco 2023), while the attitudes observed in India present a stark contrast. Research indicates a notably positive orientation toward technological adoption in the Indian context (Roy 2024). Our paper is a first step in exploring how these differences affect attitudes, opinions and beliefs and how these in turn, shape cultural artefacts such as literary novels.

To explore this further, we look at technology, particularly information technology (IT) including modern communication technology, in novels. The impact of technology varies considerably depending on geographical, economic and demographic factors, especially when comparing developed and developing countries and populations with different age distributions. In this case, our study conducts a comparative analysis of representations of IT in a corpus of Indian and German novels published between 2000 and 2024. In our study, we refer to IT as the use of technological and electronic devices to collect, store and disseminate information, which includes both hardware and software and other relevant components and functions. Thus, our primary research is: How do Indian English and German literary novels differ in their depiction of IT-related technologies? To operationalize this we measure the frequency of mentions of IT-related terms in German and Indian English novels.

Modelling which factors influence the variable of interest, the frequency of mentions of IT-related objects and practices, is complicated. There are probably two factors that directly influence our variable of interest: attitudes towards technology and literary conventions about the inclusion of technology (and other mundane objects) in literary texts. The difference in age is one of the variables influencing the attitude towards technology, other variables are traditions of public criticism of technology, ideals of the perfect life, etc., and all these variables interact (see Fig. 1).

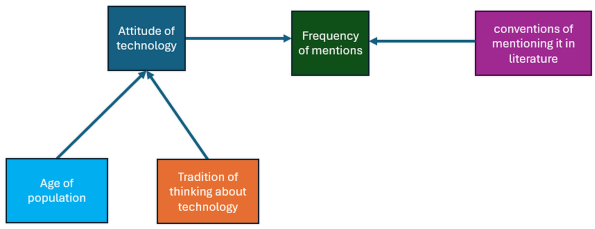

Figure 1: a simplified model of the factors determining the frequency of references to IT in novels.

Related work

Computational literary studies (CLS) have seen significant development in recent years. For German literature, computational methods have been extensively employed to analyze themes, characters, and settings (Hettinger et al., 2015; Erlin, 2017; Zehe et al., 2017; Radak et al., 2024). In contrast, while digital humanities have emerged as a growing field in India, marked by the establishment of programs, events, and academic courses, computational literary studies focusing on Indian English literature remain underexplored, with only a few notable contributions (Shanmugapriya, Menon, and Sutton, 2022; Rizi, 2023). Comparative studies of literature using CLS are even more scarce. Although some recent works (e.g., Fabo, Boisnard, and Schneider, 2024) highlight emerging trends in cross-literary computational analyses, no studies have yet been undertaken for German novels and Indian English novels. This research seeks to address this gap by providing insights into how these two literary traditions depict technology, reflecting their respective cultural, demographic, and societal contexts.

Methods

We created a corpus of forty Indian English and German novels.[2] We randomly selected 40 names of novels published between 2000 and 2024 by renowned authors. The Indian English novels were selected for both canonical and non-canonical novels by authors, including those that won or were nominated for literary awards (Appendix 1). The German novels were selected from a larger list of novels by authors who have won or been nominated for literary awards (Appendix 2).

For the analysis, we chose two approaches: a wordlist-based approach and an approach using a large language model.

- For the word-list approach, we had to curate the terms relevant to IT. Handcrafted lists are usually too small and idiosyncratic, so we applied a widely used approach: starting with one word or a short list of manually selected seed words and expanding this list by adding similar words. To do this, we used fastText word embeddings in English and German to collect the list of terms in each language. Starting with the term 'computer', we selected the nearest neighbours and used them (after removing orthographic variants) as new seed words. We manually cleaned and pruned words unrelated to IT. To make sure that the wordlists weren't already biased in some way, we decided to use one list for both corpora. To create this common list, we automatically translated each list into the respective other language and merged them. As a result we had the same list in German and English. Then we extracted all sentences in each of the corpora containing IT-related references based on the word lists.
- For the LLM-based approach we segmented the novel in chunks of 300 tokens[3] and prompted ChatGPT to extract the IT-related sentences.[4]

In both cases we tokenized the texts (tokens and sentences) with Spacy (Honnibal, Montani 2017).[5]

Evaluating a method that extracts very rare features is challenging. We focused on precision and manually checked the results on one novel, Aravind Adiga's Last Man in Tower. The wordlist approach counted 193 out of 10322 sentences as hits, but only 33 of these were true positives (precision: 0.24), 160 were false positives. The LLM counted 139 sentences and 107 of them were true positives (precision: 0.77), almost all the hits of the wordlist approach were part of the hits of the LLM approach.

Results

The wordlist-based approach shows a clear and significant difference between the frequencies of IT related mentions in German and Indian English literature:

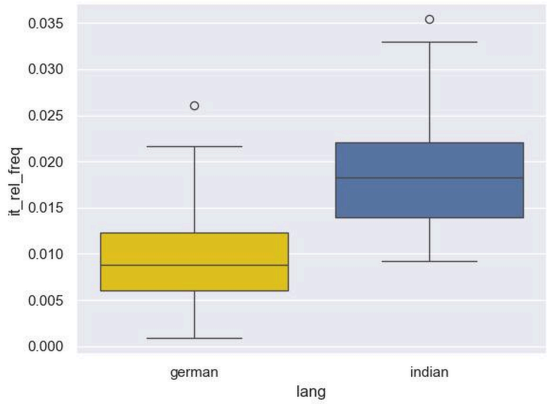

Figure 2: Comparison between the relative frequency of sentences containing IT related references in 40 German and Indian English novels based on a wordlist sentence extraction.

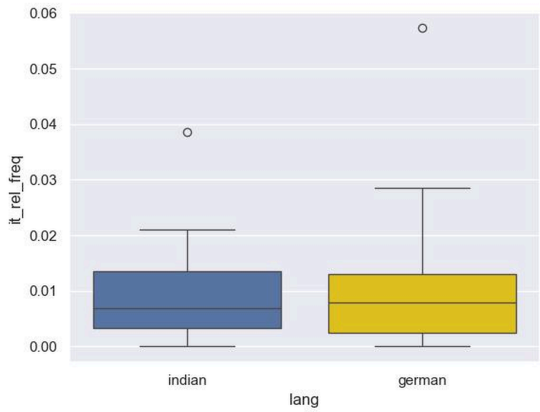

Fig. 3: Comparison between the relative frequency of sentences containing IT related references in 40 German and Indian English novels based on a modern large language model..

Discussion

Our results show that there is no difference in the frequency of IT-related references between the Indian English and German novels, which is not consistent with our initial hypothesis that a different age population and different attitudes towards technology are reflected in literary texts. The wordlist based method, which showed more promising results, is severely hampered by the fact that many terms have multiple meanings in different contexts, e.g. 'notebook', 'drive', 'chips', 'click' etc.

Our results must be seen as very preliminary, as even this first small step of a comparative cross-cultural corpus analysis was complicated by numerous problems that arose in the process. The comparability of the corpora is problematic due to the different literary traditions and conventions. Cross-linguistic comparative analysis is complicated by the very different linguistic conditions. For example, many IT terms in German are English loanwords.

Appendix 1: The corpus of Indian English novels

Alka Joshi	The Henna Artist	2020
Anita Desai	The Artist of Disappearance	2011
Anjum Hasan	History's Angel	2023
Aravind Adiga	Last Man in Tower	2011
Aravind Jayan	Teen Couple Have Fun Outdoors	2022
Aravind Adiga	The White Tiger	2008
Avni Doshi	Girl in White Cotton	2019
Shauna Singh Baldwin	The Selector of Souls	2012
Deepti Kapoor	Age of Vice	2023
Hamid. Mohsin	The Reluctant Fundamentalist	2007
Jeet Thayil	Narcopolis	2012
Manu Joseph	Serious Men	2010
Manu Joseph	The Illicit Happiness of Other People	2012
Harini Nagendra	The Bangalore Detectives Club	2022
Kiran Desai	The Inheritance of Loss	2006
Krupa Ge	What We Know about Her	2021
V Sanjay Kumar	The Third Squad: A Noir Novel	2017
Karan Mahajan	The Association of Small Bombs	2016
Amrita Mahale	Milk Teeth	2018
Saikat Majumdar	The Scent of God	2019
Megha Majumdar	A Burning	2020
Roliinton Mistry'	Family Matters	2001
Kiran Nagarkar	Jasoda: A Novel	2017
Anita Nair	Mistress	2005
Prajwal Parajuly	Land Where I Flee	2013
Jerry Pinto	Em and the Big Hoom	2012
Rajasree Variyar	The Daughters of Madurai: A Novel	2023
Arundhati Roy	The Ministry' of Utmost Happiness	2017
Arundhati Roy	Sleeping On Jupiter	2015
Bina Shah	Before She Sleeps	2018

Hansda Sowvendra Shekhar	The Adivasi Will Not Dance: Stories	2015
Preeti Shenoy	A Hundred Little Flames	2017
Tanuj Solanki	Neon Noon	2016
Amitav Ghosh	Hungry Tiger	2004
Shubhangi Swarup	Latitudes of Longing	2018
Tishani Doshi	The Pleasure Seekers	2010
Upamanyu Chatterjee	The Mammaries of the Welfare State	2000
Varun Thomas Mathew	The Black Dwarves of the Good Little Bay	2019
Padma Venkatraman	The Bridge Home	2019
Madhuri Vijay	The Far Field: A Novel	2019

Appendix 2: The corpus of German novels

oschmann, Marion	Schwarzweißroman	2005
Steinfest, Heinrich	Ein dickes Fell: Chengs dritter Fall	2006
Zaimoglu, Feridun	Leyla	2006
Hacker, Katharina	Die Habenichtse	2006
Gruber, Sabine	Über Nacht	2007
Steinaecker, Thomas von	Wallner beginnt zu fliegen	2007
Timm, Uwe	Halbschatten	2008
Bärfuss, Lukas	Hundert Tage	2008
Setz, Clemens J.	Die Frequenzen	2009
Merkel, Rainer	Lichtjahre entfernt	2009
Köhlmeier, Michael	Idylle mit ertrinkendem Hund	2010
Abonji, Melinda Nadj	Tauben fliegen auf	2010
Rabinovici, Doron	Andernorts	2010
Streeruwitz, Marlene	Die Schmerzmacherin	2011
Genazino, Wilhelm	Wenn wir Tiere wären	2011
Bronsky, Alina	Die schärfsten Gerichte der tatarischen Küche	2012
Thome, Stephan	Fliehkräfte	2012
Setz, Clemens J.	Indigo	2012
Schulz, Frank	Onno Viets und der Irre vom Kiez	2012
Zeiner, Monika	Die Ordnung der Sterne über Como	2013
Meyer, Clemens	Im Stein	2013
Leutenegger, Gertrud	Panischer Frühling	2014
Nawrat, Matthias	Unternehmer	2014
Köhlmeier, Michael	Zwei Herren am Strand	2014
Helle, Heinz	Eigentlich müssten wir tanzen	2015
Kehlmann, Daniel	F	2013
Richter, Peter	89/90	2015
Müller, Herta	Atemschaukel	2009
Veremej, Nellja	Berlin liegt im Osten	2013
Peltzer, Ulrich	Das bessere Leben	2015

Fatah, Sherko	Das dunkle Schiff	2008
Steinfest, Heinrich	Der Allesforscher	2014
Ziegelwagner, Michael	Der aufblasbare Kaiser	2014
Stanišić, Saša	Herkunft	2019
Randt, Leif	Allegro Pastell	2020
Elmiger, Dorothee	Aus der Zuckerfabrik	2020
Ravik, Antje	Blaue Frau	2021
Schubert, Helga	Vom Aufstehen	2021
Aydemir, Fatma	Dschinns	2022
Edelbauer, Raphaela	Die Inkommensurablen	2023

Appendix 3: LLM Prompt

You are a very careful reader and annotator of text.

Extract from the following text all sentences which contain direct or indirect references to modern IT like computers, smartphones, video games, the internet etc.

Return the sentences with IT references, each one starting with *. If there are no such sentences in the text, just replay with 'NONE'.

Only extract sentences where you are certain that they contain IT references. Don't add any other text.

Notes

1. United Nations. "Germany: Average age of the population from 1950 to 2100* (median age in years)." Chart. October 17, 2024. Statista. Accessed December 08, 2024.
United Nations. "India: Average age of the population from 1950 to 2100 (median age in years)." Chart. October 2, 2024. Statista. Accessed December 08, 2024.
2. For copyright reasons we cannot publish the novels we analysed.
3. Preliminary tests showed that larger chunks resulted in less good results. On a side note, contrary to the assumption that recent models are less brittle and less irritable by small changes in the prompt, we noticed that the best results were achieved with prompts with only a very few line breaks.
4. Model: gpt-4o-mini-2024-07-18, prompt see appendix 3.
5. Our scripts and additional resources can be found here: https://github.com/fotisj/ind_ger.

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